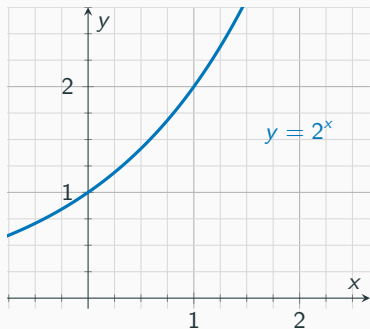


Warm-up: Estimating instantaneous rate of change

Below is the graph of $f(x) = 2^x$.



Discussion Question

1. Use the graph to estimate the value of $f'(1)$.
2. Discuss: How could you use your calculator to check your estimate?

Poll Question

Based on the graph, which of the following is true?

A $f'(1) < 2$

B $f'(1) = 2$

C $f'(1) > 2$

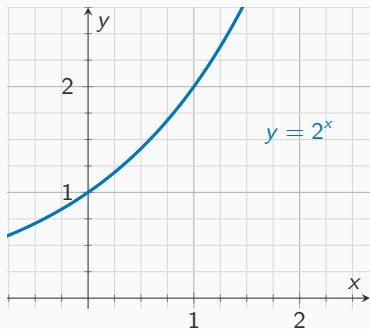
Lecture 7 The Derivative Function

Big Questions

- Can the instantaneous rate of change be seen a function?
- What can we learn from the instantaneous rate of change?

Warm-up: Estimating instantaneous rate of change

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The derivative

The instantaneous rate of change of a function f also goes by another name.

Definition

The **derivative of f at a** , written $f'(a)$, is defined to be the instantaneous rate of change of f at the point a .

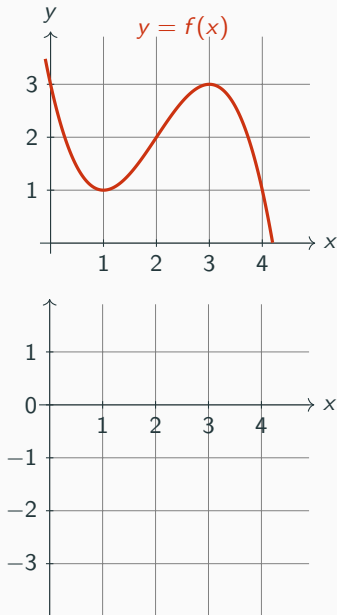
Poll Question

For each of the following, is the quantity described equal to $f'(a)$?

Options: ☒ A Yes ☐ B No

- the instantaneous rate of change of f at a
- the average rate of change of f at a
- the slope of the secant line to the graph $y = f(x)$ at $(a, f(a))$
- the slope of the tangent line to the graph $y = f(x)$ at $(a, f(a))$
- the slope of the graph $y = f(x)$ at $x = a$
- the derivative of f at $x = a$

Sketching the graph of the derivative



Discussion Question

1. Use the graph of f on the left to help you fill in the table below with approximate values.

a	0	1	2	3	4
$f'(a)$					

2. Use the values in your table to help you sketch an approximate graph of f' on the left.

The derivative function

On the previous slide, we used the graph of a function to produce a graph of the function's derivative.

So, derivatives themselves must be **functions**.

Definition

The **derivative** of $f(x)$, denoted $f'(x)$, is the **function** which returns the instantaneous rate of change of f at x (if it exists).

We can use graphs to help us visualize the derivative:

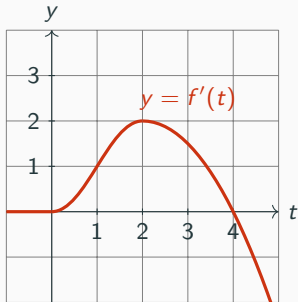
<https://www.desmos.com/calculator/kl8qrnoqxn>

After Class

At home, try to work through the following:

1. Use the c slider to move the orange dot around.
2. What does the orange dot represent?
3. Play around with desmos! You can change the function $f(x)$ and use this to sketch different derivatives in the space below. Try the following functions:
 - $f(x) = x$
 - $f(x) = x^2$
 - $f(x) = e^x$

Mr. Bean is driving along a straight road running west-east. The position y (in km) of his car east of home at time t in minutes since noon is given by $y = f(t)$. The graph of the **derivative**, $y = f'(t)$, is sketched below.



Poll Question

At what time t did Mr. Bean stop and turn around?

- A** $t = 0$ **B** $t = 2$
C $t = 4$ **D** Never

Discussion Question

Write a story that could represent Mr. Bean's journey.

What does the sign of $f'(x)$ tell us?

Poll Question

With your group, discuss how to complete each of the following statements. Be prepared to share your reasoning with the class!

(a) If $f'(x) > 0$ on an interval, then, on that interval, $f(x)$ must be:

A positive **B** increasing **C** concave up **D** other

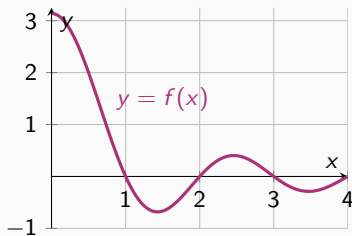
(b) If $f'(x) < 0$ on an interval, then, on that interval, $f(x)$ must be:

A negative **B** decreasing **C** concave down **D** other

(c) If $f'(x) = 0$ on an interval, then $f(x)$ is on that interval.

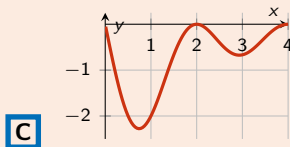
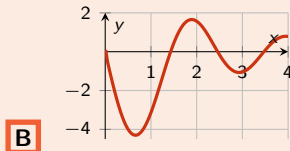
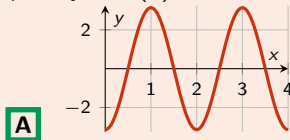
Geometric properties of the derivative

Consider the function whose graph is shown below.



Poll Question

Which of the following shows the graph of $y = f'(x)$?



To do before next class:

- Please read Section 2.2.
- Complete questions 4–7 from the Week 2 Checkpoint.
- Complete questions 12 - 17 in Homework 2.

Have a great day!